



Data Analytics 2-2. Probability

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Probabilistic model

- **Modeling**
 - Approximate reality with a simple(mathematical) model
 - Experiment: Flip two coins
 - Observation(Outcomes): (Head, Head)
 - All observations: $\{(H, H), (H, T), (T, H), (T, T)\}$
 - The goal: Building up a probabilistic model for an experiment with observations

Probabilistic model

- **Probabilistic model; 확률모형**

- Assign a number to each outcome or a set of outcomes
- Mathematical description of an uncertain situation
- Elements
 - The sample space Ω : the set of all possible outcomes of the experiment
 - The event space $\mathcal{A}$: the space of potential results of the experiment
 - The probability p : measuring the probability or degree of belief that the event will occur
 - ✓ $p(A)$: Probability of an event A , where an event $A \in \mathcal{A}$
- Probability space $(\Omega, \mathcal{A}, p)$

Probabilistic model

- **Probabilistic model**

- **Example.** Flipping two coins
- The sample space $\Omega = \{(H, H), (H, T), (T, H), (T, T)\}$
- Examples of events: $(H, H), (H, T), (T, H), (T, T)$
- The event space $\mathcal{A}$: The collection of subsets of Ω (often, the power set of Ω)
- The probability p of an event $A = (H, H)$: $p(A) = 0.25$

Probabilistic model

- **Random variable; 확률변수**

- In reality, many outcomes are numeric, e.g., stock prices
- Mathematically, a random variable is a function

$$X: \Omega \rightarrow \mathbb{R}$$

that takes an element of Ω and returns a numerical value

- Random variable: X , numerical value: x
- We can associate an event that a random variable X has the value x , i.e.,

$$\{w \in \Omega \mid X(w) = x\}$$

- Generally,

$$p_X(S) = p(X \in S) = p(\{w \in \Omega \mid X(w) \in S\})$$

Probabilistic model

- **Random variable**

- Example. Flipping two coins

- What if random variable X is the number of heads?

$$X((H, H)) = 2, X((H, T) \text{ or } (T, H)) = 1, X((T, T)) = 0$$

- The probability p of an event $A = (H, H)$

$$p(A) = p(X = 2) = 0.25$$

Discrete and Continuous Probabilities

- **Discrete random variable; 이산확률변수**

- The values that a random variable X takes is discrete, e.g., 0, 1, 2, ...
- Probability mass function(PMF)
 - The probability that a discrete random variable X takes a particular value x

$$p_X(x) = p(X = x) = p(\{w \in \Omega \mid X(w) = x\})$$

- Examples: Bernoulli, Uniform, Binomial, Poisson, Geometric, and so forth

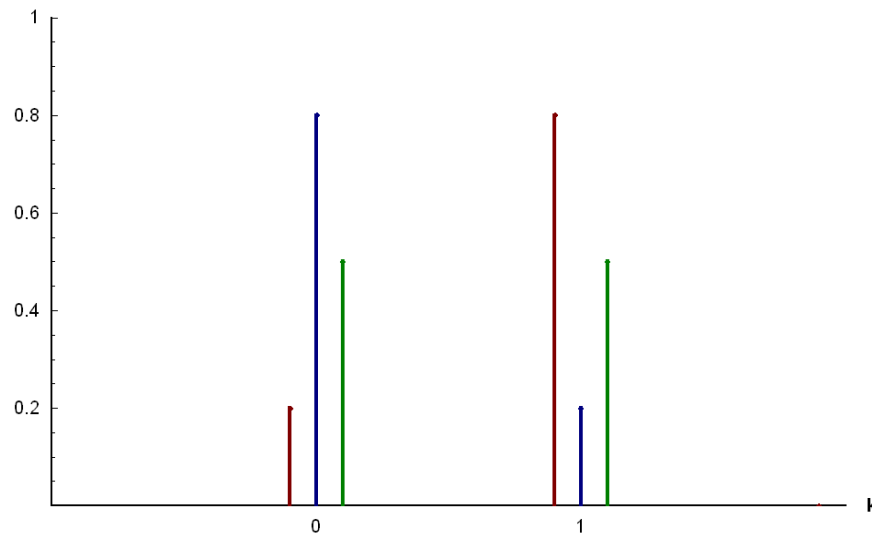
Discrete and Continuous Probabilities

- **Bernoulli X with parameter $p \in [0, 1]$; Bernoulli(p)**

- A random variable can take only binary value, i.e., 0, 1
- The probability mass function p_X

$$p_X(k; p) = \begin{cases} p & k = 1 \\ 1 - p & k = 0 \end{cases}$$

- $E[X] = p$, $\text{var}[X] = p(1 - p)$



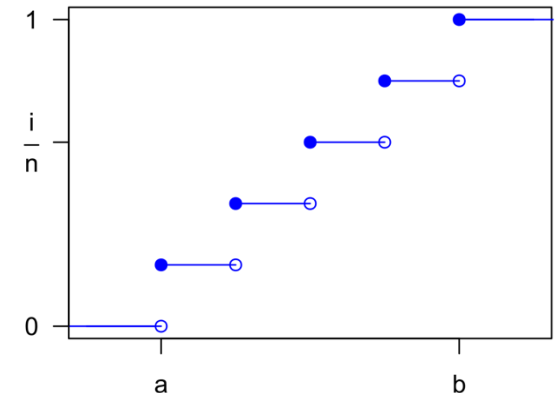
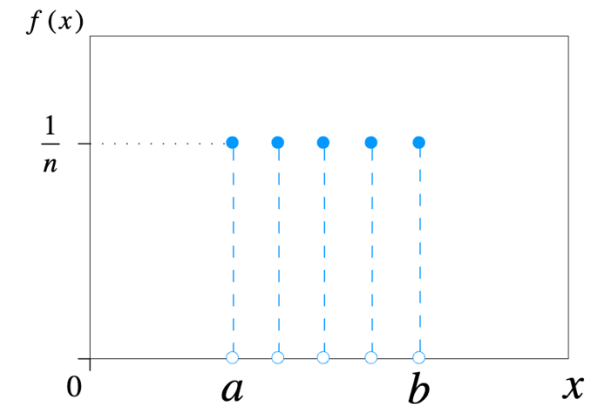
Discrete and Continuous Probabilities

- **Uniform X with parameter a, b ; Uniform(a, b)**

- Integers a, b , where $a \leq b$
- Choose a number of $\Omega = \{a, a + 1, \dots, b\}$ uniformly at random
- The probability mass function p_X

$$p_X(k; a, b) = \frac{1}{b - a + 1}$$

- $E[X] = \frac{a+b}{2}$, $\text{var}[X] = \frac{(b-a+1)^2 - 1}{12}$

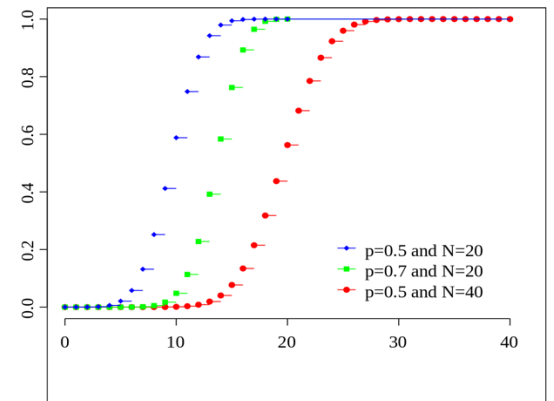
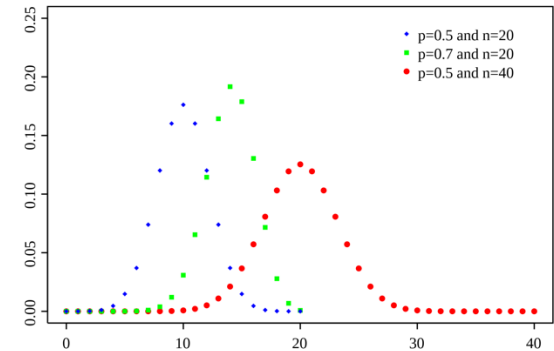


Discrete and Continuous Probabilities

- **Binomial X with parameter n, p ; Binomial(n, p)**
 - Modeling the number of successes in a given number of independent trials
 - The probability mass function p_X for k successes in n independent Bernoulli trials

$$p_X(k; n, p) = \binom{n}{p} p^k (1 - p)^{n-k}$$

- $E[X] = np$, $\text{var}[X] = np(1-p)$



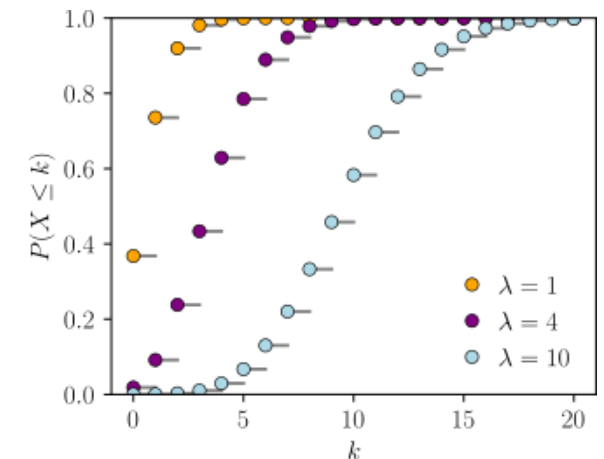
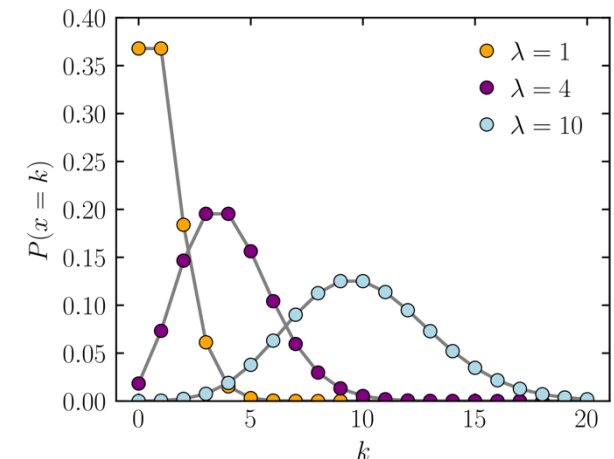
Discrete and Continuous Probabilities

- **Poisson X with parameter λ ; Poisson(λ)**

- Binomial(n, p): Modeling the number of successes in n independent trials with success probability p
- Very large n and very small p , such that $np = \lambda$
- The probability mass function p_X

$$p_X(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- $E[X] = \lambda$, $\text{var}[X] = \lambda$



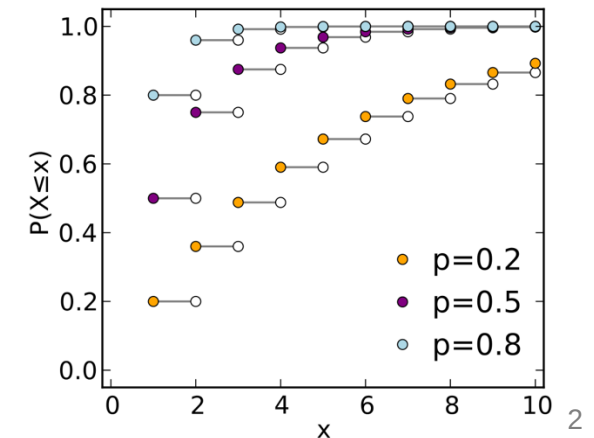
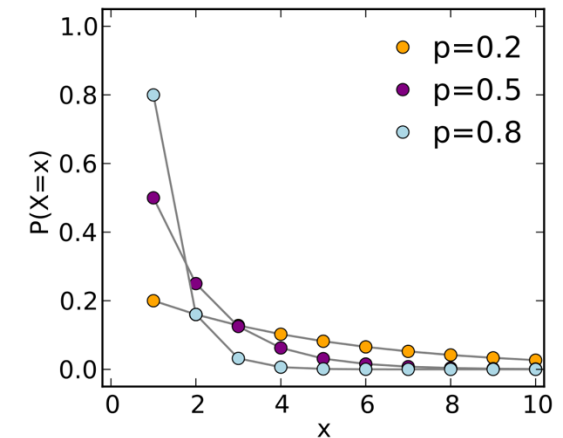
Discrete and Continuous Probabilities

- **Geometric X with parameter p ; Geometric(p)**

- Experiment: infinitely many independent Bernoulli trials, where each trial has success probability p
- Random variable X : number of trials until the first success
- The probability mass function p_X

$$p_X(k; p) = (1 - p)^{k-1}p$$

- $E[X] = \frac{1}{p}$, $\text{var}[X] = \frac{1-p}{p^2}$



Discrete and Continuous Probabilities

- **Discrete random variable**

- **Example.** Flipping two coins

- Probability mass function

$$p_X(x) = p(X = x) = \begin{cases} \frac{x+1}{4} & , & x = 0, 1 \\ \frac{x-1}{4} & , & x = 2 \end{cases}$$

- Discrete probability distribution

x	0	1	2
$p(X = x)$	1/4	2/4	1/4

Discrete and Continuous Probabilities

- **Joint PMF and Marginal PMF**

- Joint PMF

- For two random variables X, Y , consider two events $\{X = x\}$ and $\{Y = y\}$, then

$$p_{X,Y}(x, y) = p(\{X = x\}, \{Y = y\})$$

- Sum of joint probabilities: $\sum_x \sum_y p_{X,Y}(x, y) = 1$

- Marginal PMF

- Given two random variables, only single random variable is considered

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Discrete and Continuous Probabilities

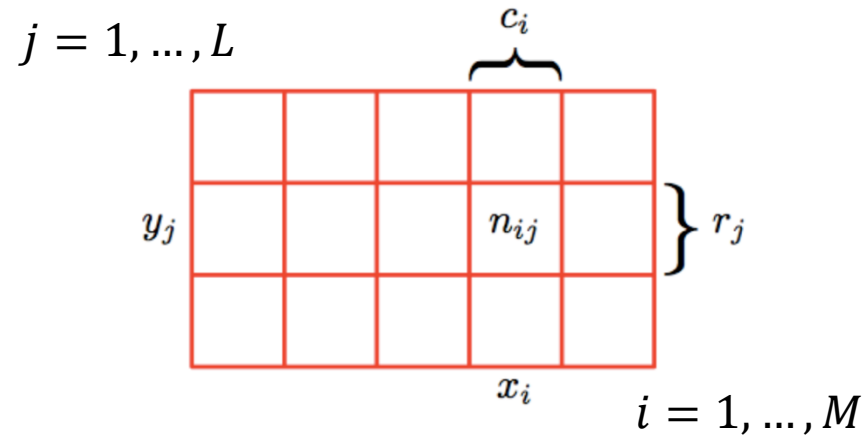
- **Joint PMF and Marginal PMF**

- **Example.** Joint and marginal distributions of a pair of discrete random variables X, Y

	x_1	x_2	x_3	x_4	$p_Y(y)$
y_1	$4/32$	$2/32$	$1/32$	$1/32$	$8/32$
y_2	$3/32$	$6/32$	$3/32$	$3/32$	$15/32$
y_3	$9/32$	0	0	0	$9/32$
$p_X(x)$	$16/32$	$8/32$	$4/32$	$4/32$	$32/32$

Discrete and Continuous Probabilities

- Joint PMF and Marginal PMF



- Joint probability

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

- Marginal probability

$$p(X = x_i) = \frac{c_i}{N}, \quad p(Y = y_j) = \frac{r_j}{N}$$

Discrete and Continuous Probabilities

- **Continuous random variable; 연속확률변수**
 - The values that a random variable X takes is continuous
 - Probability density function(PDF) f_X
 - A function $f_X: \mathbb{R}^D \rightarrow \mathbb{R}$ if $\forall x \in \mathbb{R}^D: f_X(x) \geq 0$, its integral exists, and

$$\int_{\mathbb{R}^D} f_X(x) dx = 1$$

- For example,

$$p(a \leq X \leq b) = \int_a^b f_X(x) dx$$

Discrete and Continuous Probabilities

- **Continuous random variable; 연속확률변수**

- PMF vs. PDF

- PMF

$$p(a \leq X \leq b) = \sum_{x:a \leq x \leq b} p_X(x), \quad p_X(x) \geq 0, \quad \sum_x p_X(x) = 1$$

- PDF

$$p(a \leq X \leq b) = \int_a^b f_X(x) dx, \quad f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1$$

Discrete and Continuous Probabilities

- **Exponential X with parameter $\lambda > 0$; exponential(λ)**
 - Modeling the probability of the distance between events in a Poisson point process
 - e.g., a process in which events occur continuously and independently at a constant average rate

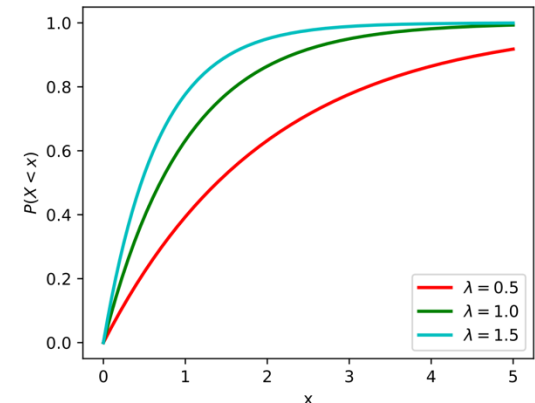
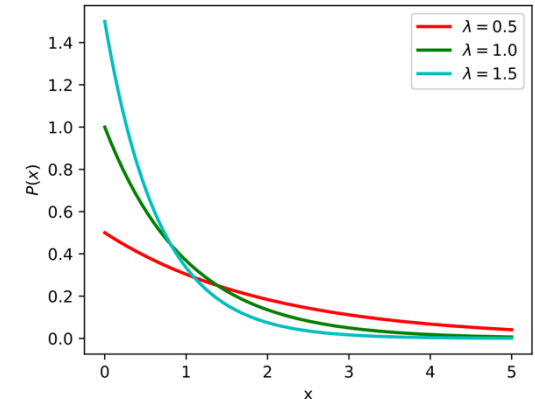
- The probability density function f_X

$$f_X(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- The cumulative distribution function F_X

$$F_X(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

- $E[X] = 1/\lambda$, $\text{var}[X] = 1/\lambda^2$



Discrete and Continuous Probabilities

- **Joint PDF and Marginal PDF**

- Joint PDF

- For two continuous random variables X, Y and subset B of the two dimensional plane,

$$p((X, Y) \in B) = \iint_{(x,y) \in B} f_{X,Y}(x, y) dx dy$$

- Our particular interest: $B = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$
 - Sum of joint probabilities: $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$

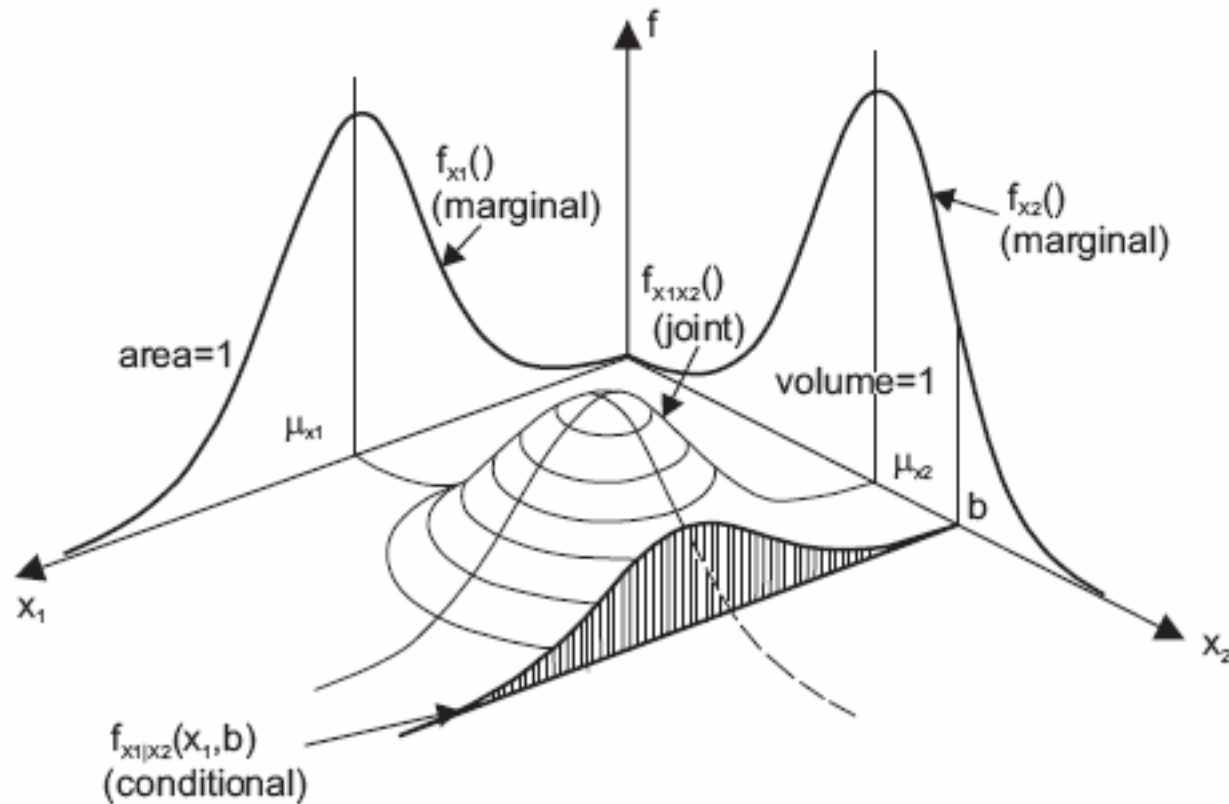
- Marginal PDF

- Given two random variables, only single random variable is considered

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$$

Discrete and Continuous Probabilities

- Joint PDF and Marginal PDF



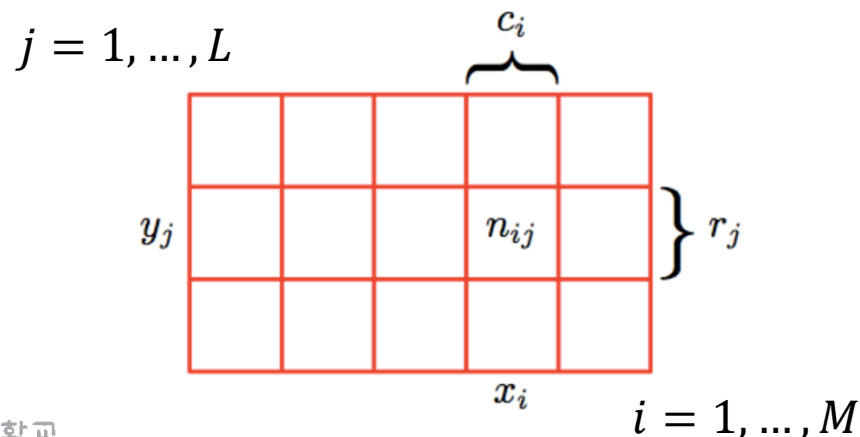
Sum Rule, Product Rule, and Bayes' Theorem

- **Sum Rule**

- Two events that could occur, but cannot occur at the same time

$$p(x) = \begin{cases} \sum_{y \in Y} p_{X,Y}(x, y) & \text{if } y \text{ is discrete} \\ \int_{y \in Y} f_{X,Y}(x, y) dx & \text{if } y \text{ is continuous} \end{cases}$$

- Example.



$$p(X = x_i) = \frac{c_i}{N} = \frac{1}{N} \sum_{j=1}^L n_{ij}$$

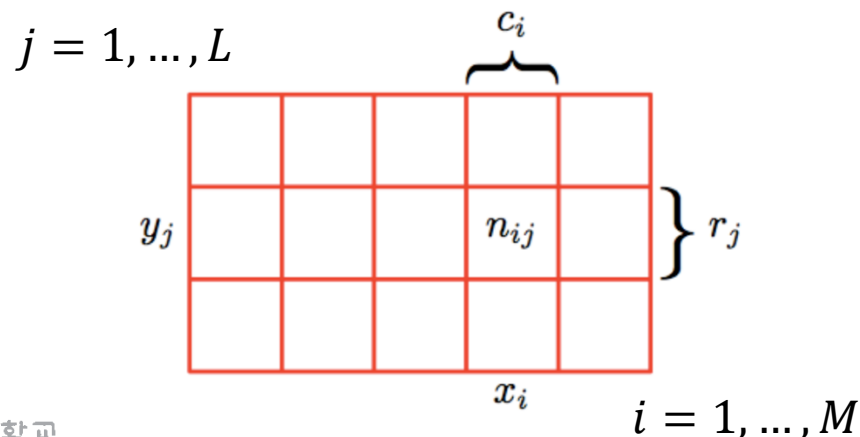
Sum Rule, Product Rule, and Bayes' Theorem

- **Conditional Probability**

- A measure of the probability of an event occurring given that another event is already known to have occurred
- The conditional probability of x given y : $p(y | x)$

$$p(y | x) = \frac{p(x, y)}{p(x)}$$

- Example.



$$p(Y = y_i | X = x_i) = \frac{n_{ij}}{c_i}$$
$$p(X = x_i | Y = y_i) = \frac{n_{ij}}{r_j}$$

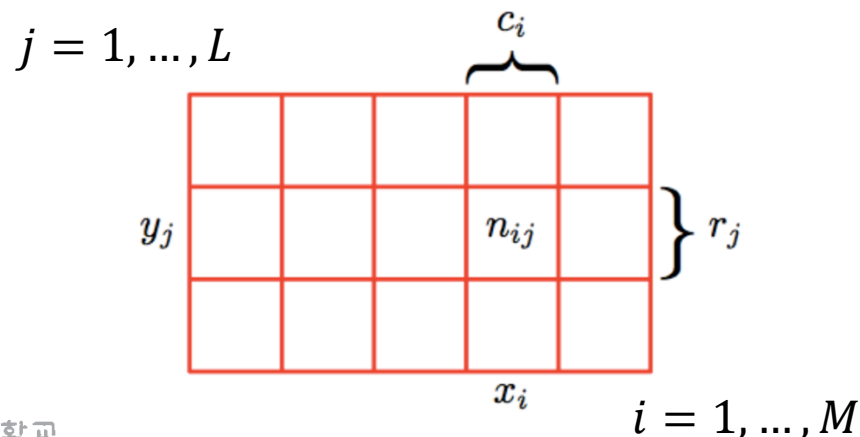
Sum Rule, Product Rule, and Bayes' Theorem

- **Product Rule**

- Calculating probability of the intersection of events using conditional probability

$$p(x, y) = p(x) \cdot p(y|x) = p(y) \cdot p(x|y)$$

- Joint = marginal of the first $\times$ conditional of the second given the first
- Joint = marginal of the second $\times$ conditional of the first given the second
- Example.



$$\begin{aligned} p(X = x_i, Y = y_j) &= \frac{n_{ij}}{N} \\ &= \frac{n_{ij}}{c_i} \cdot \frac{c_i}{N} = \frac{n_{ij}}{r_j} \cdot \frac{r_j}{N} \end{aligned}$$

Sum Rule, Product Rule, and Bayes' Theorem

- **Bayes' theorem**

- The probability of an event based on prior knowledge of conditions
- x : state/cause/original value $\rightarrow$ y : result/resulting action/noisy measurement
- Model: $p(x)$ (prior) and $p(y|x)$ (cause $\rightarrow$ result)
- Inference: $p(x|y)$

$$\underbrace{p(x|y)}_{\text{posterior}} = \frac{\overbrace{p(y|x)}^{\text{likelihood}} \cdot \overbrace{p(x)}^{\text{prior}}}{\underbrace{p(y)}_{\text{evidence}}}$$

Sum Rule, Product Rule, and Bayes' Theorem

- **Bayes' theorem**

$$\underbrace{p(x|y)}_{\text{posterior}} = \frac{\overbrace{p(y|x)}^{\text{likelihood}} \cdot \overbrace{p(x)}^{\text{prior}}}{\underbrace{p(y)}_{\text{evidence}}}$$

- $p(F)$: *prior probability*
- $p(F|E)$: *posterior probability*
- **확률에 대한 관점 (동전의 앞면이 나올 확률이 50%?)**
 - 전통적인 관점 : 빈도주의 (100번 동전을 던졌을 때 50번은 앞면이 나온다)
 - 베이지안 주의 관점 : 동전의 앞면이 나왔다는 주장의 신뢰도가 50%
- **핵심 개념 : 새로운 정보를 토대로 어떤 사건이 발생했다는 주장에 대한 신뢰도를 갱신**

Sum Rule, Product Rule, and Bayes' Theorem

- Applying Bayes' Theorem
 - **Examples.** Suppose that one person in 100,000 has a particular disease. There is a test for the disease that gives a positive result 99% of the time when given to someone with the disease. When given to someone without the disease, 99.5% of the time it gives a negative result. Find
 - a) the probability that a person who test positive has the disease.
 - b) the probability that a person who test negative does not have the disease.

Sum Rule, Product Rule, and Bayes' Theorem

- Applying Bayes' Theorem

- **Examples.** Suppose that one person in 100,000 has a particular disease. There is a test for the disease that gives a positive result 99% of the time when given to someone with the disease. When given to someone without the disease, 99.5% of the time it gives a negative result. Find the probability that a person who test positive has the disease.

- Let D be the event that the person has the disease, and E be this person tests positive.

- $p(D) = 0.00001; p(\bar{D}) = 1 - p(D) = 0.99999$

- $p(E|D) = 0.99; p(\bar{E}|D) = 1 - p(E|D) = 0.01$

- $p(\bar{E}|\bar{D}) = 0.995; p(E|\bar{D}) = 1 - p(\bar{E}|\bar{D}) = 0.005$

- $$p(D|E) = \frac{p(E|D)p(D)}{p(E|D)p(D) + p(E|\bar{D})p(\bar{D})} = \frac{0.99 \cdot 0.00001}{0.99 \cdot 0.00001 + 0.005 \cdot 0.99999} \approx 0.002$$

Sum Rule, Product Rule, and Bayes' Theorem

- Applying Bayes' Theorem

- **Examples.** Suppose that one person in 100,000 has a particular disease. There is a test for the disease that gives a positive result 99% of the time when given to someone with the disease. When given to someone without the disease, 99.5% of the time it gives a negative result. Find the probability that a person who test negative does not have the disease.

- Let D be the event that the person has the disease, and E be this person tests positive.

- $p(D) = 0.00001; p(\bar{D}) = 1 - p(D) = 0.99999$

- $p(E|D) = 0.99; p(\bar{E}|D) = 1 - p(E|D) = 0.01$

- $p(\bar{E}|\bar{D}) = 0.995; p(E|\bar{D}) = 1 - p(\bar{E}|\bar{D}) = 0.005$

- $$p(\bar{D}|\bar{E}) = \frac{p(\bar{E}|\bar{D})p(\bar{D})}{p(\bar{E}|\bar{D})p(\bar{D}) + p(\bar{E}|D)p(D)} = \frac{0.995 \cdot 0.99999}{0.995 \cdot 0.99999 + 0.01 \cdot 0.00001} \approx 0.9999999$$

Summary Statistics and Independence

- **Expectation and Variance; 기대값과 분산**

- Describing properties of probability distributions
- Expectation of PMF/PDF

$$E[X] = \sum_x x p_X(x), \quad E[X] = \int_x x f_X(x) dx$$

- Variance(Spread) of PMF/PDF

$$\text{var}[X] = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

– Standard deviation $\sigma_X = \sqrt{\text{var}[X]}$

Summary Statistics and Independence

- **Expectation and Variance; 기대값과 분산**
 - Properties

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

$$\text{var}[aX + b] = a^2 \text{var}[X]$$

Summary Statistics and Independence

- **Expectation and Variance; 기대값과 분산**
 - **Examples.** What is the expectation and variance of the random variable X , where X is the number that comes up when a fair die is rolled?

$$E(X) = 1/6 * (1+2+3+4+5+6) = 7/2$$

$$E(X^2) = 1/6 * (1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = 91/6$$

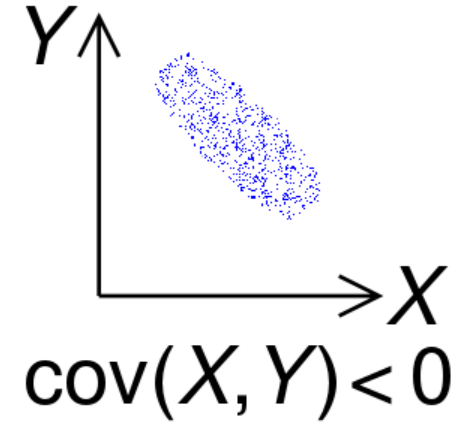
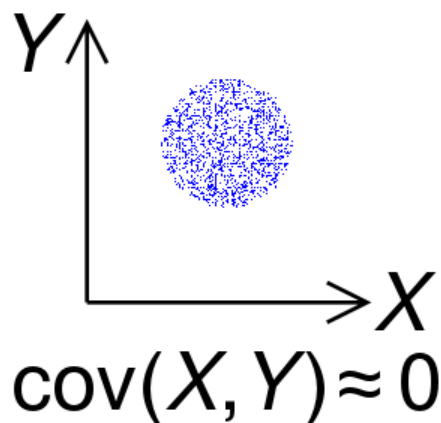
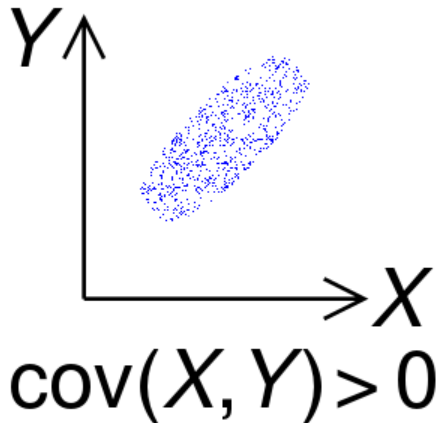
$$E(X) = 7/2$$

$$V(X) = 91/6 - 49/4 = 35/12$$

Summary Statistics and Independence

- **Covariance; 공분산**

- A measure of the joint variability of two random variables
- Given two random variables X and Y , quantifying the degree of dependence
 - Positive (If $X \uparrow$, $Y \uparrow$) and Negative (If $X \uparrow$, $Y \downarrow$)



Summary Statistics and Independence

- **Covariance; 공분산**

- For two jointly random variables X and Y , the covariance is defined as the expected value of the product of their deviations from their individual expected values:

$$\begin{aligned} \text{cov}(X, Y) &= E[(X - E[X]) \cdot (Y - E[Y])] \\ &= E[XY - XE[Y] - YE[X] + E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] - E[X]E[Y] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y] \end{aligned}$$

- When $\text{cov}(X, Y) = 0$,
 - We say that X and Y have an orthogonal relationship?
 - We say that X and Y are uncorrelated?

Summary Statistics and Independence

- **Covariance; 공분산**

- Properties

- $cov(X, X) = 0$

- $cov(aX + b, Y) = E[(aX + b)Y] - E[aX + b]E[Y] = aE[XY] - aE[X]E[Y] = acov(X, Y)$

- $cov(X, Y + Z) = E[X(Y + Z)] - E[X]E[Y + Z]$
 $= E[XY] + E[XZ] - E[X]E[Y] - E[X]E[Z] = cov(X, Y) + cov(X, Z)$

- $var(X + Y) = E[(X + Y)^2] - (E[X + Y])^2$
 $= E[X^2] + 2E[XY] + E[Y^2] - ((E[X])^2 + 2E[X]E[Y] + (E[Y])^2)$
 $= (E[X^2] - (E[X])^2) + (2E[XY] - 2E[X]E[Y]) + (E[Y^2] - (E[Y])^2)$
 $= var(X) + 2cov(X, Y) + var(Y)$

Summary Statistics and Independence

- **Correlation coefficients; 상관계수**
 - Normalizing the covariance by dividing by the geometric mean of the total variances for the two random variables
 - Transforming the covariance into the value between -1 and 1

$$\begin{aligned}\rho(X, Y) &= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = E \left[\frac{(X - E[X])}{\sigma_X} \cdot \frac{(Y - E[Y])}{\sigma_Y} \right] \\ &= \frac{E[XY] - E[X]E[Y]}{\sqrt{E[X^2] - (E[X])^2} \sqrt{E[Y^2] - (E[Y])^2}}\end{aligned}$$

Summary Statistics and Independence

- **Extension to Multivariate**
 - Consider a random vector $\mathbf{x}$ and $\mathbf{y}$ as follows,

$$\mathbf{x} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \in \mathbb{R}^n, \quad \mathbf{y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \in \mathbb{R}^n$$

- Expectation and covariance for multivariate random variables

$$E[\mathbf{x}] = \begin{bmatrix} E[X_1] \\ \vdots \\ E[X_n] \end{bmatrix}, \quad cov(\mathbf{x}, \mathbf{y}) = E[\mathbf{x}\mathbf{y}^T] - E[\mathbf{x}]E[\mathbf{y}]^T \in \mathbb{R}^{n \times m}$$

Summary Statistics and Independence

- **Extension to Multivariate**

- Variance for multivariate random variables: $var(\mathbf{x}) = cov(\mathbf{x}, \mathbf{x}) \in \mathbb{R}^{n \times n}$

$$\Sigma_{\mathbf{x}} = var(\mathbf{x}) = \begin{bmatrix} cov(X_1, X_1) & cov(X_1, X_2) & \cdots & cov(X_1, X_n) \\ cov(X_2, X_1) & cov(X_2, X_2) & \cdots & cov(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ cov(X_n, X_1) & cov(X_n, X_2) & \cdots & cov(X_n, X_n) \end{bmatrix}$$

Summary Statistics and Independence

- **Independence**

- Two random variables X, Y are statistically independent if and only if

$$p(x, y) = p(x)p(y)$$

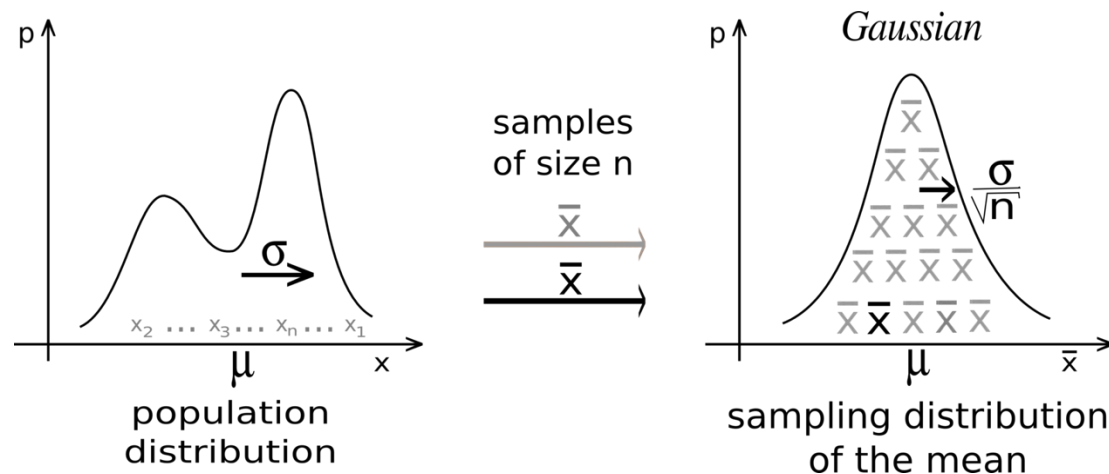
- $p(y|x) = p(y)$
- $p(x|y) = p(x)$
- $\text{var}[x + y] = \text{var}[x] + \text{var}[y]$
- $\text{cov}[x, y] = 0$

Gaussian Distribution

- **Normal (also called Gaussian) Random Variable**

- Central limit theorem; 중심극한정리

- Suppose $X_1, X_2, \dots$ is a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{var}[X_i] = \sigma^2$
- Then, as n approaches infinity, the distribution of $\bar{X}_n$ gets close to the normal distribution with mean μ and variance σ^2/n



Gaussian Distribution

- **Normal (also called Gaussian) Random Variable**

- General Normal $\mathcal{N}(\mu, \sigma^2)$

- The probability density function $f_X(x)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

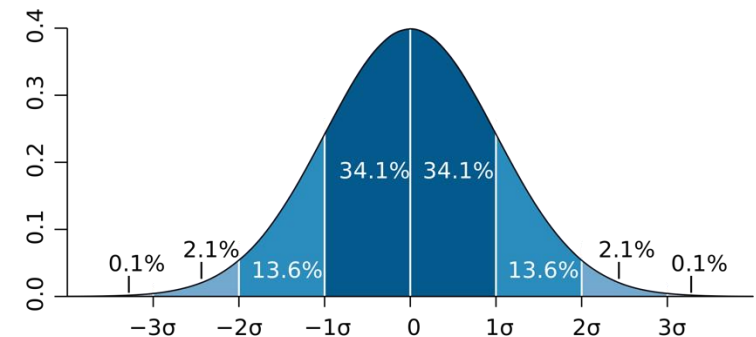
- $E[X] = \mu, \text{var}[X] = \sigma^2$

- Standard Normal $\mathcal{N}(0,1)$

- The probability density function $f_X(x)$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2)$$

- $E[X] = 0, \text{var}[X] = 1$



Next Session

- Event Logs



Any questions?

